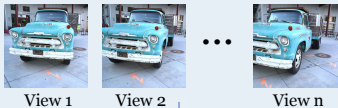
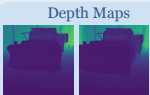


## (a) Spatial Memory

Context RGB (observed)



DA3 Geometry



Depth Scale

Align Pose

Projection

Global Spatial Memory

Render



Validity Mask

Update Memory

## (b) Wan2.2 VACE Generative Prior

RGB Inputs



RGB & Mask

RGB

VACE Control Branch

Residuals

WAN2.2 DiT

B0

B5

B10

B15

Projector + Zero Conv

Projector + Zero Conv

Projector + Zero Conv

Projector + Zero Conv

## (c) Geometric Latent Diffusion

Depth Anything 3

Input Features

Level 1 Feature

Noisy Tensor

plücker rays

Generative Prior 1

GP2

GP3

GP4

Enc.

3D Attention

AdaLN

Dec.

3D Attention

AdaLN

Attention Block

Attention Block

Attention Block

L0 Feature

L1 conditioned cascade

L1 Feature

DA3 Block 8-11

L2 Feature

L3 Feature

RGB head

Geometry head

